PCA and Latent Factor Models (I)

MFE5340: AI in FE: Quantitative Investment

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- The excess return of asset i at time t can be represent by the following linear model:

$$R_{it} = \boldsymbol{\beta}_i^\top \boldsymbol{f}_t + e_{it},$$

where $K \times 1$ vector $\boldsymbol{\beta}_i$ represent factor exposures of asset i, $K \times 1$ vector $\boldsymbol{f}_t$ represents the excess returns of those K factors at time t, and e_{it} is the zero-mean idiosyncratic error, such that $\mathbb{E}[e|\boldsymbol{f}] = 0$.

- Taking the covariance of both sides gives:

$$\boldsymbol{\Sigma}_r = \boldsymbol{\beta}\boldsymbol{\Sigma}_f\boldsymbol{\beta}^\top + \boldsymbol{\Sigma}_e,$$

where $\boldsymbol{\Sigma}_r$ and $\boldsymbol{\Sigma}_f$ are the covariance matrices of assets and factors, respectively. $\boldsymbol{\Sigma}_e$ is the covariance matrix of the error terms.

While factor models are elegant, their challenge lies in the identification of factors. There are two main approaches in this regard and both have gained a lot of attention.

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Two Approaches

- The first approach assumes either factors or factor exposures are **known** and **observable**. For example, Fama and French factors.

- The second approach assumes both factors and factor exposures are **latent**. Its starting point is the covariance matrix equation:

$$\boldsymbol{\Sigma}_r = \boldsymbol{\beta}\boldsymbol{\Sigma}_f\boldsymbol{\beta}^\top + \boldsymbol{\Sigma}_e,$$

  It implies that the covariance of the factors should be able to explain a large portion of the assets' co-movement. We estimate the latent factors and their corresponding exposures from $\boldsymbol{\Sigma}_r$.

- Solution: Principal Components Analysis (PCA).

> By identifying and retaining the principal components that capture the most variance, PCA helps in simplifying the data structure, making it easier to identify underlying risk factors that drive asset returns.

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2 Principal Components Analysis

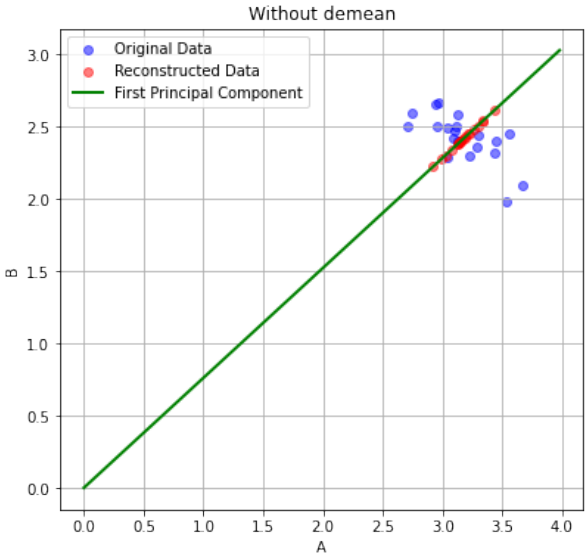
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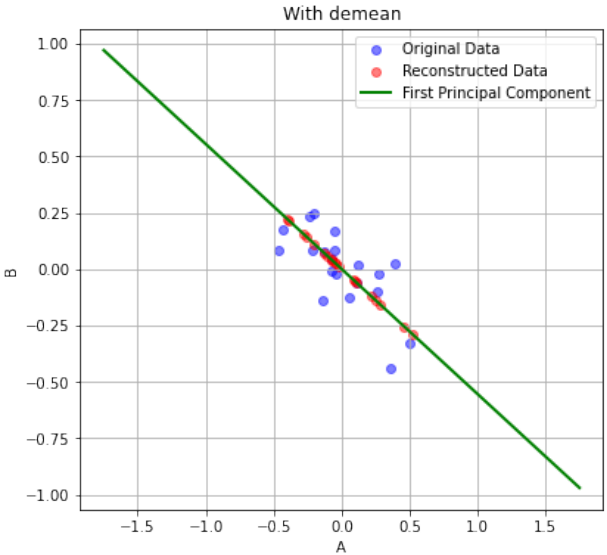
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PCA is a powerful statistical technique used to uncover the underlying structure in a dataset by transforming it into **a new coordinate system**. This transformation highlights the directions (principal components) along which the data varies the most.

- Before applying PCA, it is crucial to **center the data** by subtracting the mean of each variable (i.e., demean).
- PCA can be viewed as **a rotational transformation** of the data. It identifies the directions in which the data points exhibit the largest variance from the center.

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- Players: N assets with $N \times 1$ random vector $\boldsymbol{r} = [r_1, \cdots, r_N]^\top$ representing the returns of these assets for a given period, and $N \times N$ matrix $\boldsymbol{\Sigma}_r$ being the covariance matrix.
- **We aim to find a linear combination of these N assets, i.e., $f_1 = \boldsymbol{w}_1^\top \boldsymbol{r}$, such that the variance of f_1 (i.e., var(f_1)) is maximized.**

> **The constructed f_1 has a clear intuitive explanation: it can be viewed as a portfolio of the original N assets weighted by $\boldsymbol{w}_1$, and this portfolio is viewed as a factor.**

- However, we can always change var(f_1) by scaling up $\boldsymbol{w}_1$. Hence, we need to impose the following constraint: $\boldsymbol{w}_1^\top \boldsymbol{w}_1 = 1$.

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- The optimization problem:

$$\max \boldsymbol{w}_1^\top \boldsymbol{\Sigma}_r \boldsymbol{w}, \quad \text{subject to} \quad \boldsymbol{w}^\top \boldsymbol{w} = 1.$$

- The Lagrange function ($\mathcal{L}$):

$$\mathcal{L} = \boldsymbol{w}_1^\top \boldsymbol{\Sigma}_r \boldsymbol{w}_1 - \lambda_1 (\boldsymbol{w}_1^\top \boldsymbol{w}_1 - 1).$$

- First-order condition (with respect to $\boldsymbol{w}_1$):

$$\boldsymbol{\Sigma}_r \boldsymbol{w}_1 = \lambda_1 \boldsymbol{w}_1.$$

$\boldsymbol{w}_1$ is an eigenvector of $\boldsymbol{\Sigma}_r$, and λ_1 is the corresponding eigenvalue. Since we want to **maximize** $\text{var}(f_1)$, $\boldsymbol{w}_1$ corresponds to the eigenvector of $\boldsymbol{\Sigma}_r$ with the **largest** eigenvalue (i.e., 1st component). $f_1 = \boldsymbol{w}_1^\top \boldsymbol{r}$ is the portfolio of these assets with the largest variance.

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1. The first principal component of $\boldsymbol{r}$ is the linear combination (factor) $f_1 = \boldsymbol{w}_1^\top \boldsymbol{r}$ that maximizes $\text{var}(f_1)$ subject to $\boldsymbol{w}_1^\top \boldsymbol{w}_1 = 1$.

2. The second principal component of $\boldsymbol{r}$ is the linear combination (factor) $f_2 = \boldsymbol{w}_2^\top \boldsymbol{r}$ that maximizes $\text{var}(f_2)$ subject to $\boldsymbol{w}_2^\top \boldsymbol{w}_2 = 1$ and being orthogonal to $\boldsymbol{w}_1$.

3. The k-th principal component of $\boldsymbol{r}$ is the linear combination (factor) $f_k = \boldsymbol{w}_k^\top \boldsymbol{r}$ that maximizes $\text{var}(f_k)$ subject to $\boldsymbol{w}_k^\top \boldsymbol{w}_k = 1$ and being orthogonal to $\boldsymbol{w}_j$ for $j = 1, \cdots, k-1$.

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- Given that these factors are obtained through PCA, they are often referred to as *principal component factors* (PC factors) or *statistic factors* in academic literature.
- For factor k, the exposure of the assets $\boldsymbol{\beta}_k = [\beta_{1k}, \beta_{2k}, \cdots, \beta_{Nk}]^\top$ equals $\boldsymbol{w}_k = [w_{1k}, w_{2k}, \cdots, w_{Nk}]^\top$.
- Here is why:

$$\beta_{ik} = \frac{\text{cov}(r_i, f_k)}{\text{var}(f_k)} = \frac{\text{cov}(r_i, \sum_j w_{jk} r_j)}{\lambda_k} = \frac{\sum_j w_{jk} \text{cov}(r_i, r_j)}{\lambda_k}.$$

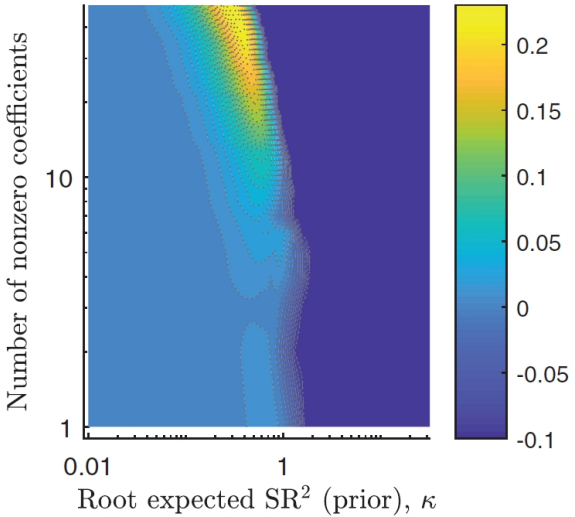
According to the definition of eigenvectors:

$$\boldsymbol{\Sigma}_r \boldsymbol{w}_k = \lambda_k \boldsymbol{w}_k \Rightarrow \text{for each row } i, \sum_j w_{jk} \text{cov}(r_i, r_j) = \lambda_k w_{ik}.$$

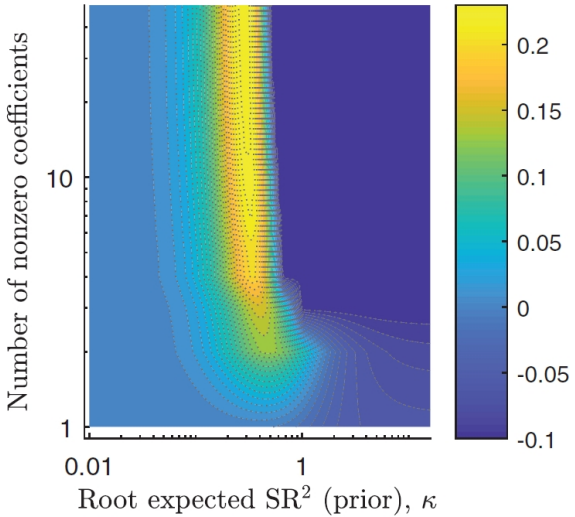
Therefore

$$\beta_{ik} = \frac{\lambda_k w_{ik}}{\lambda_k} = w_{ik}, \quad i = 1, \cdots, N.$$

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(a) Raw 50 anomaly portfolios



(b) PCs of 50 anomaly portfolios

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Not without Limit!

- A factor return series $\boldsymbol{f}$ with a non-zero mean implies that we can use the average returns of factors (i.e., the estimated factor risk premia) for asset pricing.

- From the perspective of PCA's motivation, when constructing these factors, we only utilized the second **central** moment information of the original data relative to its mean (i.e., variance), **without explicitly using the first moment information.**

- Besides, we have constructed the PC factors solely based on asset return data, **completely ignoring other information related to returns, such as firm characteristics.** Is it possible to incorporate them into the PCA framework? The answer is yes.

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- PCA aims to explain as much variation of the original data (i.e., the structure of $\mathbf{\Sigma}_r$) as possible using a few principal components (i.e., a few factors). We often select only the first K components.

- PCA achieves **dimensionality reduction**, and we are effectively using K principal components to perform a **low-rank approximation** of the original data:

$$\tilde{\mathbf{R}} = \tilde{\mathbf{F}}_K \mathbf{\Lambda}_K^\top + \mathbf{E},$$

  where $T \times N$ matrix $\tilde{\mathbf{R}}$ represents the mean-centered return, $N \times K$ matrix $\mathbf{\Lambda}_K$ consists of the first K eigenvectors, $T \times K$ matrix $\tilde{\mathbf{F}}_K = \tilde{\mathbf{R}}\mathbf{\Lambda}_K$ is the factor returns, and $\mathbf{E}$ represents the error term.

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- View $\tilde{\mathbf{R}} = \tilde{\mathbf{F}}_K \mathbf{\Lambda}_K^\top + \mathbf{E}$ as a linear regression model.
- We can solve for $\tilde{\mathbf{F}}_K = \tilde{\mathbf{R}} \mathbf{\Lambda}_K (\mathbf{\Lambda}_K^\top \mathbf{\Lambda}_K)^{-1}$ by **minimizing** $\|\mathbf{E}\|_F^2$ (where $\|.\|_F^2$ is the Frobenius norm). When $\mathbf{\Lambda}_K$ consists of the first K eigenvectors, $\mathbf{\Lambda}_K^\top \mathbf{\Lambda}_K = \mathbf{I}_K$, and $\tilde{\mathbf{F}}_K = \tilde{\mathbf{R}} \mathbf{\Lambda}_K (\mathbf{\Lambda}_K^\top \mathbf{\Lambda}_K)^{-1} = \tilde{\mathbf{R}} \mathbf{\Lambda}_K$.
- Among all design matrices, when the columns of $\mathbf{\Lambda}_K$ are the first K eigenvectors, the resulting low-rank approximation $\tilde{\mathbf{F}}_K \mathbf{\Lambda}_K$ is the **best** low-rank approximation (see lecture note).

This allows us to view PCA from a least squares perspective. This is very important!

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- Performing PCA is equivalent to minimizing the error, i.e., $\|\mathbf{E}\|_F^2$, of the following linear regression model:

$$\tilde{\mathbf{R}} = \tilde{\mathbf{F}}_K \mathbf{\Lambda}_K^\top + \mathbf{E}.$$

  We can view PCA as solving a linear regression problem using the least squares method.

- In traditional PCA, we first derive $\mathbf{\Lambda}_K$ by eigen-decomposition, and then compute $\tilde{\mathbf{F}}_K$ via $\tilde{\mathbf{F}}_K = \tilde{\mathbf{R}}\mathbf{\Lambda}_K(\mathbf{\Lambda}_K^\top \mathbf{\Lambda}_K)^{-1}$.

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- Performing PCA is equivalent to minimizing the error, i.e., $\|\mathbf{E}\|_F^2$, of the following linear regression model:

$$\tilde{\mathbf{R}} = \tilde{\mathbf{F}}_K \mathbf{\Lambda}_K^\top + \mathbf{E}.$$

  We can view PCA as solving a linear regression problem using the least squares method.

- In traditional PCA, we first derive $\mathbf{\Lambda}_K$ by eigen-decomposition, and then compute $\tilde{\mathbf{F}}_K$ via $\tilde{\mathbf{F}}_K = \tilde{\mathbf{R}}\mathbf{\Lambda}_K(\mathbf{\Lambda}_K^\top \mathbf{\Lambda}_K)^{-1}$.

What if we don't know $\mathbf{\Lambda}_K$?

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Alternating Least Squares (ALS): an **iterative** optimization technique commonly used for solving least squares problems. In the PCA context, we can leverage ALS to iteratively update the factor loadings $\mathbf{\Lambda}_K$ and the factor returns $\tilde{\mathbf{F}}_K$ until convergence.

- Both $\mathbf{\Lambda}_K$ and $\tilde{\mathbf{F}}_K$ are unknown and initialize them randomly.
- Alternately update $\mathbf{\Lambda}_K^{LS}$ and $\tilde{\mathbf{F}}_K^{LS}$ (superscript LS indicates results derived from ALS) **by solving the least squares problem**:
  - Fix $\mathbf{\Lambda}_K^{LS}$ and solve for $\tilde{\mathbf{F}}_K^{LS}$.
  - Fix $\tilde{\mathbf{F}}_K^{LS}$ and solve for $\mathbf{\Lambda}_K^{LS}$.
- PCA provides the best low-rank approximation for $\tilde{\mathbf{R}}$. If the ALS iterations lead to the global optimal solution, then $\mathbf{\Lambda}_K^{LS}$ and $\tilde{\mathbf{F}}_K^{LS}$ will be consistent with those obtained by PCA.

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- When updating $\mathbf{\Lambda}_K^{LS}$ given $\tilde{\mathbf{F}}_K^{LS}$, its formula

$$\mathbf{\Lambda}_K^{LS} = \tilde{\mathbf{R}}^\top \tilde{\mathbf{F}}_K^{LS} \left[(\tilde{\mathbf{F}}_K^{LS})^\top (\tilde{\mathbf{F}}_K^{LS}) \right]^{-1}.$$

  does not guarantee that the resulting $\mathbf{\Lambda}_K^{LS}$ will be orthogonal.

- Solution: An orthogonalization step can be added after each iteration of $\mathbf{\Lambda}_K^{LS}$. This step transforms the columns of $\mathbf{\Lambda}_K^{LS}$ into **a set of orthogonal bases** that span the column space of $\mathbf{\Lambda}_K^{LS}$.

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- When updating $\tilde{\mathbf{F}}_K^{LS}$ given $\mathbf{\Lambda}_K^{LS}$ (after orthogonalization), its formula

$$\tilde{\mathbf{F}}_K^{LS} = \tilde{\mathbf{R}}\mathbf{\Lambda}_K^{LS}\left[(\mathbf{\Lambda}_K^{LS})^\top(\mathbf{\Lambda}_K^{LS})\right]^{-1}.$$

  does not ensure that the diagonal elements of the covariance matrix of $\tilde{\mathbf{F}}_K^{LS}$ are ordered from largest to smallest (or even that the covariance matrix of $\tilde{\mathbf{F}}_K^{LS}$ is diagonal).

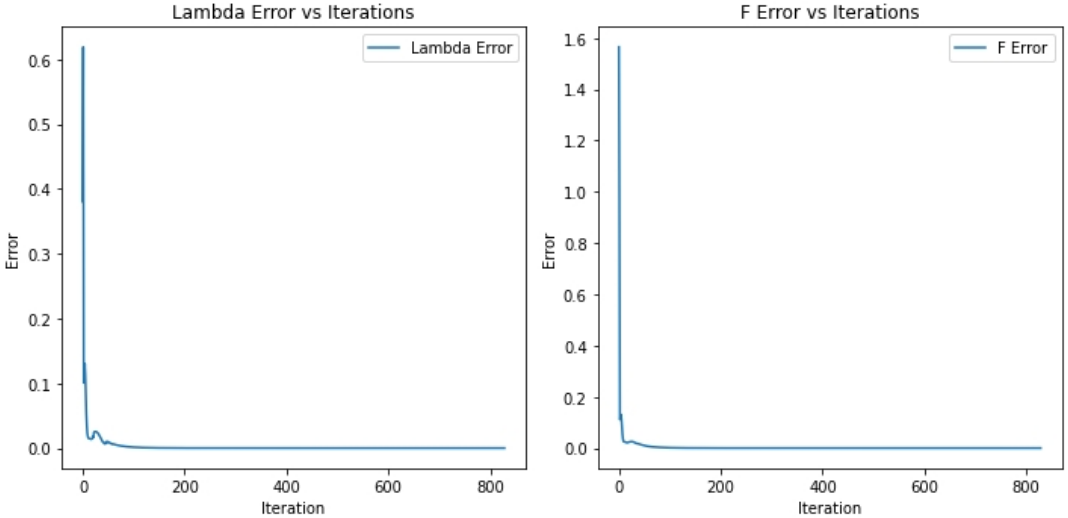
- Reason: $(\tilde{\mathbf{F}}_K^{LS})(\mathbf{\Lambda}_K^{LS})^\top$ possesses **rotational invariance**, meaning they are identifiable only up to a linear rotation.

- Why? For any orthogonal matrix $\mathbf{Q}$, the following holds:

$$(\tilde{\mathbf{F}}_K^{LS}\mathbf{Q})(\mathbf{\Lambda}_K^{LS}\mathbf{Q})^\top = \tilde{\mathbf{F}}_K^{LS}\mathbf{Q}\mathbf{Q}^\top(\mathbf{\Lambda}_K^{LS})^\top = (\tilde{\mathbf{F}}_K^{LS})(\mathbf{\Lambda}_K^{LS})^\top.$$

- Solution: Rotate $\tilde{\mathbf{F}}_K^{LS}$ after each iteration by eigen-decomposition on its covariance matrix.

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An Example


```
Lambda from Direct PCA:
[[-0.29084986 -0.39253192 -0.36073389  0.66168226 -0.11292247  0.38096323]
 [ 0.07017416 -0.48522419 -0.51178797 -0.09354403  0.11947739 -0.67203061]
 [ 0.17530103 -0.41772319  0.49646502  0.23344993  0.15801193 -0.12871698]
 [ 0.6652759  -0.24625328 -0.19552011 -0.13555483  0.14984088  0.52414312]
 [-0.21346332 -0.54980017  0.3426491  -0.41964008  0.19416384  0.1742704 ]
 [-0.44723986 -0.21631302  0.16745526 -0.13853312 -0.38939636  0.08118065]
 [ 0.43614119 -0.13821442  0.25137548  0.17941709 -0.72900251 -0.20142286]
 [ 0.03674576  0.0846644   0.33852503  0.50306301  0.45252606 -0.18551595]]

Lambda from Iterative Method:
[[-0.29084986 -0.39253192 -0.36073389  0.66168226 -0.11292247  0.38096323]
 [ 0.07017416 -0.48522419 -0.51178797 -0.09354403  0.11947739 -0.67203061]
 [ 0.17530103 -0.41772319  0.49646502  0.23344993  0.15801193 -0.12871698]
 [ 0.6652759  -0.24625328 -0.19552011 -0.13555483  0.14984088  0.52414312]
 [-0.21346332 -0.54980017  0.3426491  -0.41964008  0.19416384  0.1742704 ]
 [-0.44723986 -0.21631302  0.16745526 -0.13853312 -0.38939636  0.08118065]
 [ 0.43614119 -0.13821442  0.25137548  0.17941709 -0.72900251 -0.20142286]
 [ 0.03674576  0.0846644   0.33852503  0.50306301  0.45252606 -0.18551595]]

Lambda Error after rotation: 7.96674933644124e-12
F Error: 8.528792087366218e-11

Covariance of F (Direct PCA):
[0.0953114  0.09090247 0.08728113 0.08476512 0.08224683 0.07881684]

Covariance of F (Iterative Method):
[0.0953114  0.09090247 0.08728113 0.08476512 0.08224683 0.07881684]
```


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- There are scenarios where ALS becomes more attractive or necessary.
- Recent developments in PCA methodology have introduced modifications that do not have analytical solutions and therefore require iterative approaches like ALS. One notable example is the instrumented PCA (IPCA) method proposed by Kelly, Pruitt, and Su (2019):
  - given estimate of $\mathbf{\Gamma}$, we have $\boldsymbol{f}_{t+1} = \left(\mathbf{\Gamma}^\top \mathbf{Z}_t^\top \mathbf{Z}_t \mathbf{\Gamma}\right)^{-1} \mathbf{\Gamma}^\top \mathbf{Z}_t^\top \boldsymbol{r}_{t+1}$;
  - given estimates of the series $\{\mathbf{f}_t\}, t = 1, \cdots, T-1$, we have

$$\text{vec}(\mathbf{\Gamma}^\top) = \left(\sum_{t=1}^{T-1} \mathbf{Z}_t^\top \mathbf{Z}_t \otimes \boldsymbol{f}_{t+1} \boldsymbol{f}_{t+1}^\top\right)^{-1} \left(\sum_{t=1}^{T-1} \left[\mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}^\top\right]^\top \boldsymbol{r}_{t+1}\right).$$

  where $\otimes$ is Kronecker product.

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- Let the $T \times N$ matrix $\mathbf{R}$ represent asset returns. Besides, let $N \times 1$ vector $\bar{\boldsymbol{R}}$ represent the sample mean returns of N assets; and $\boldsymbol{R}_t$ represent the **transpose** of the t-th row of $\mathbf{R}$ (so $\boldsymbol{R}_t$ itself is a $N \times 1$ column vector!).

- Sample covariance $\mathbf{S}$ (need derivation?):

$$\mathbf{S} = \frac{1}{T}\sum_{t=1}^{T}\begin{bmatrix} (R_{1t}-\bar{R}_1)(R_{1t}-\bar{R}_1) & (R_{1t}-\bar{R}_1)(R_{2t}-\bar{R}_2) & \cdots & (R_{1t}-\bar{R}_1)(R_{Nt}-\bar{R}_N) \\ (R_{2t}-\bar{R}_2)(R_{1t}-\bar{R}_1) & (R_{2t}-\bar{R}_2)(R_{2t}-\bar{R}_2) & \cdots & (R_{2t}-\bar{R}_2)(R_{Nt}-\bar{R}_N) \\ \vdots & \vdots & \ddots & \vdots \\ (R_{Nt}-\bar{R}_N)(R_{1t}-\bar{R}_1) & (R_{Nt}-\bar{R}_N)(R_{2t}-\bar{R}_2) & \cdots & (R_{Nt}-\bar{R}_N)(R_{Nt}-\bar{R}_N) \end{bmatrix}$$

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- First expression:

$$\mathbf{S} = \frac{1}{T}\sum_{t=1}^{T}(\boldsymbol{R}_t - \bar{\boldsymbol{R}})(\boldsymbol{R}_t - \bar{\boldsymbol{R}})^\top.$$

- Second expression:

$$\mathbf{S} = \frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}},$$

  where $\tilde{\mathbf{R}} = \mathbf{R} - \mathbf{1}_T\bar{\boldsymbol{R}}^\top$ is the **centered** (or **demeaned**) return matrix. This is the second expression of $\mathbf{S}$, which is in matrix form.

- Third expression:

$$\mathbf{S} = \frac{1}{T}\mathbf{R}^\top\mathbf{R} - \bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top.$$

- Derivation?

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- Instead of performing eigen-decomposition to the covariance matrix $(1/T)\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$, we can also perform eigen-decomposition to the covariance matrix $(1/N)\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$.
- These two approaches can be linked by the SVD decomposition of $\tilde{\mathbf{R}}$.
- SVD of $\tilde{\mathbf{R}}$:

$$\tilde{\mathbf{R}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top,$$

where $\mathbf{U}$ is a $T \times T$ orthogonal matrix (the left singular vectors of $\tilde{\mathbf{R}}$), $\mathbf{V}$ is a $N \times N$ orthogonal matrix (the right singular vectors of $\tilde{\mathbf{R}}$), and $\mathbf{\Sigma}$ is a $T \times N$ diagonal matrix with non-negative real numbers on the diagonal, denoted as σ_i, being the singular values of $\tilde{\mathbf{R}}$.

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Connections between Eigenvalues

6 $(1/T)\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ and $(1/N)\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$

- Eigenvalues of $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$:

$$\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} = \frac{1}{T}\mathbf{V}\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma}\mathbf{V}^\top = \mathbf{V}\left(\frac{\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma}}{T}\right)\mathbf{V}^\top$$

  The eigenvalues of $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ are $\lambda_i = \frac{\sigma_i^2}{T}$.

- Eigenvalues of $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$:

$$\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top = \frac{1}{N}\mathbf{U}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^\top\mathbf{U}^\top = \mathbf{U}\left(\frac{\boldsymbol{\Sigma}\boldsymbol{\Sigma}^\top}{N}\right)\mathbf{U}^\top$$

  The eigenvalues of $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$ are $\mu_i = \frac{\sigma_i^2}{N}$.

$$\boxed{\sigma_i = \sqrt{\lambda_i T} = \sqrt{\mu_i N}.}$$

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Connections between Eigenvectors

6 $(1/T)\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ and $(1/N)\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$

- From $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} = \mathbf{V}\left(\frac{\mathbf{\Sigma}^\top\mathbf{\Sigma}}{T}\right)\mathbf{V}^\top$, it is easy to see that the columns of $\mathbf{V}$, $\boldsymbol{v}_i$, are the eigenvectors of $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$.
- Similarly, from $\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top = \mathbf{U}\left(\frac{\mathbf{\Sigma}\mathbf{\Sigma}^\top}{N}\right)\mathbf{U}^\top$, it is easy to see that the columns of $\mathbf{U}$, $\boldsymbol{u}_i$, are the eigenvectors of $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$.
- Since $\tilde{\mathbf{R}}$ can be written as $\tilde{\mathbf{R}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, it implies that $\tilde{\mathbf{R}}\boldsymbol{v}_i = \sigma_i\boldsymbol{u}_i$ for each i. Then:

$$\boldsymbol{u}_i = \frac{1}{\sigma_i}\tilde{\mathbf{R}}\boldsymbol{v}_i.$$

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Why Bother?

6 $(1/T)\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ and $(1/N)\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$

- In empirical asset pricing, we often encounter situations where the number of assets N far exceeds the number of time periods T. In such cases, performing PCA using $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ becomes not only computationally intensive but also problematic in terms of matrix rank.

- Instead, a more practical approach is to work with the smaller $T \times T$ covariance matrix $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$. By focusing on the $T \times T$ matrix, we avoid the rank-deficiency issue and maintain the accuracy of PCA while improving computation efficiency.

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Questions?

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